The root-finding decomposition oracle inside ecbench's index calculus: binary and prime ladders against rho

2026-10-13. Studies `research/pdp_root_finder_ecbench_20261013/` (binary ladder, its claim round in `claim_round/`) and `research/pkm_tower_ecbench_20261013/` (PKM's prime-field Kummer tower). Oracles: `ic_framework::rootfind` (`rootfind-mitm`, `rootfind-pairs`). Factor base for prime curves: `ic_framework::prime_tower` (`prime-kummer`). Protocols registered before the runs; one amendment (the oracle's accounting, before the binary ladder restarted). **Class: engineering.**

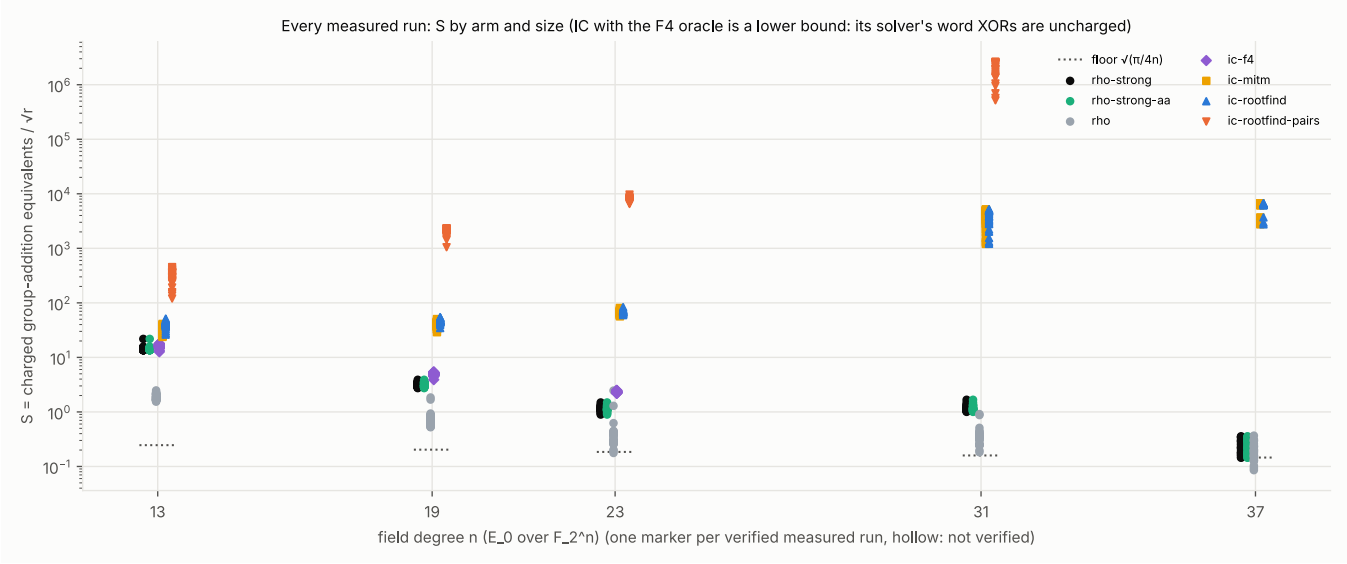
What was measured

Index calculus end to end — factor base, pair table, relations, linear algebra, verification — in ecbench's one unit, S = charged group-addition equivalents $/\sqrt{r}$, on one-target workloads, every run verified by the harness and every run replayed by the audit.

- **Binary ladder.** $E_0 : y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^n}$, $n = 13, 19, 23, 31, 37$, standard subspace base of dimension $d = 5, 7, 8, 10, 12$; arms: strong signed-Frobenius rho (reference), the inherited-F4 Gröbner oracle (`pdp3-koblitz`, $n \le 23$), the harness's `mitm:m=3` table oracle, `rootfind-mitm:m=3`, `rootfind-pairs:m=3` ($n \le 31$), an A/A rho control. Eight public targets, five rounds ($n \le 19$) or three.
- **Prime ladder.** $y^2 = x^3 + ax + b$ over $\mathbb{F}_p$, $p = c \cdot 2^{12} + 1$ of 20–28 bits, PKM's base $V = \mu_{2^t}$, $t = 8 \ldots 11$; arms: rho with the negation map (reference), `mitm`, `rootfind-mitm`, `rootfind-pairs` ($t \le 9$), the same oracle on a plain abscissa base of equal size (control), A/A. Eight targets, three rounds.
- **Claim round.** `rootfind-mitm` on a Frobenius-orbit base (`koblitz-orbit`) at $n = 23, 31$, so that a `vs_rho` claim can be built; one claim per workload.

All 1,248 binary-ladder runs, 896 prime-ladder runs and 256 claim-round runs verified $[d]G = Q$; every replay audit is `OK` with every replay identical.

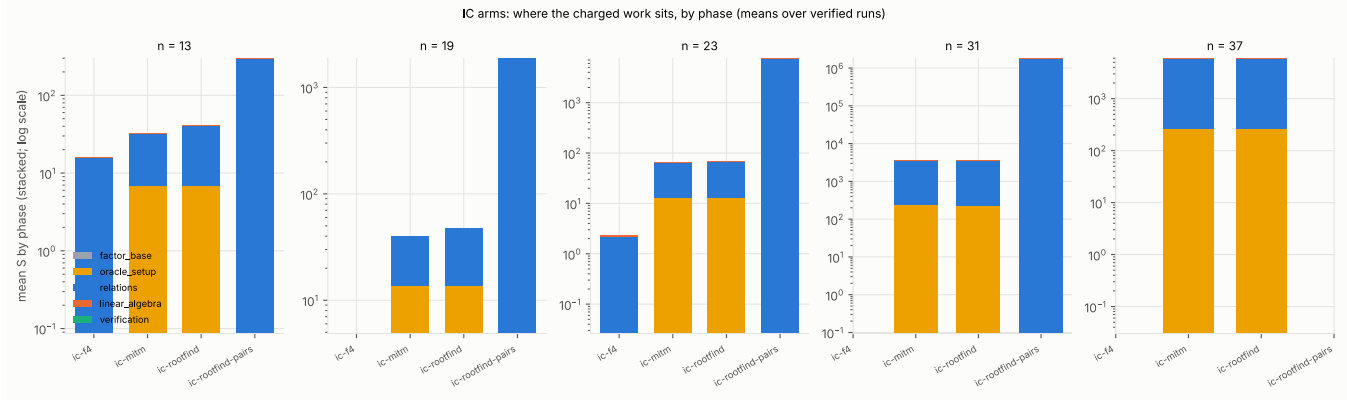
Binary ladder



Every measured run's S by arm and size, binary ladder, with the floor $\sqrt{\pi/4n}$. The F4 arm's S is a lower bound: its Gröbner word XORs are counted and not charged.

n	`rootfind` ÷ strong rho, S	÷ strong rho, wall	÷ F4, S (F4 a lower bound)	÷ F4, wall	÷ `mitm`, S	÷ `mitm`, wall
13	2.85 [2.64, 3.04]	4.18	2.56	**0.0029** (admitted)	1.263	0.953
19	15.0 [14.3, 15.7]	11.6	9.59	0.0004 (descriptive)	1.186	0.697
23	60.9 [56.1, 66.0]	34.5	29.9	0.0001 (descriptive)	1.047	0.427
31	3,053 [2,505, 3,596]	2.33	—	—	1.003	0.278
37	25,116 [20,793, 30,546]	descriptive	—	—	1.0008	0.312 (descriptive)

Paired totals ratios from `ecbench compare` with 95 % cluster-bootstrap intervals; wall ratios are the harness's admitted medians where both runs held L2. Against the F4 oracle, the whole solve is 350× faster in wall time at $n = 13$ (admitted) and about 2,500× and 10,000× at $n = 19, 23$ (descriptive; the F4 runs are too long to hold L2). Against rho, index calculus costs more on every size and the gap grows; the jump at $n = 31$ is mostly the curve's cofactor (1,492): index calculus pays for the whole group, rho only for the prime subgroup. The relations phase is 71–94 % of S, the pair table 6–29 %, factor base and linear algebra under 2 %.



Mean S by phase for the binary ladder's index-calculus arms (verified runs).

Prime ladder (PKM's Kummer tower)



Every measured run's S by arm and size, prime ladder, with the floor $\sqrt{\pi/4}$. Every arm charges every group operation.

bits of p	$\log_2 r$	`rootfind` ÷ rho, S	÷ rho, wall	tower ÷ plain base, S	`rootfind` ÷ `mitm`, S / wall
20	18.85	34.0 [27.9, 42.6]	15.5	1.006	1.21 / 1.31
22	21.19	41.9 [34.3, 53.4]	26.4	1.001	1.14 / 1.13
24	23.38	29.4 [23.7, 37.3]	22.8	1.012	1.08 / 1.17
26	23.28	96.6 [77.8, 124.1]	83.4	1.005	1.06 / 1.06
28	26.20	140.6 [113.0, 182.3]	178.6	0.999	1.03 / 1.09

Index calculus with PKM's base costs 29–141× rho with the negation map, growing by 0.26 ± 0.28 bits per bit of r (least squares over the five sizes), consistent with the $p^{1/6}$ the floors predict and too wide here to separate from zero. The tower base costs the same as a plain base of equal size within 1.2 %: its structure helps an algebraic solver, not a table. On prime curves the new oracle is 3–31 % slower than `mitm` in wall time: the prime group has no batched addition, so the block of 32 abscissae only adds overshoot.

Claims

On the binary ladder's subspace base every `vs_rho` claim build was refused: the IC1 identity names a base by its sign-and-Frobenius orbits and that base has more columns than orbits. On a Frobenius-orbit base all 16 claims build and are checked. Every claim shows index calculus slower than the strong rho on its target (online speedup 0.097–0.392 at $n = 23$, 0.007–0.020 at $n = 31$). Every claim fails the checker on independent validation (no replay receipt from a second hardware class is available in this container) and on two stage fields the claim schema requires and `ecbench claim build` does not write (`operation_accounting`, `record_class`), a builder gap independent of this measurement; nine runs held L1 instead of L2.

Decision

The root-finder replaces the Gröbner oracle inside the index-calculus pipeline at thousands of times less wall time on binary curves, matches the harness's own table oracle in charged work, and leaves the method's growth unchanged: on binary and prime curves alike, index calculus with an enumerating oracle stays above rho and the gap widens with the curve, as the pair-table floor ($p^{2/3}$ at $m = 3$) against rho's $p^{1/2}$ says it must. Moving that exponent needs a decomposition test whose per-trial cost does not grow with the factor base; the scope note `research/ic_speed_and_rho_scope_20261013/` lists the candidates and their falsifiers.

Environment: Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU, L3 260 MiB, 15 GiB RAM, kernel 6.18.44, rustc 1.97.0; `ecbench run --cpus auto` (a reserved core, other threads evicted), one thread. Conductor's control plane was unreachable in this container: no task card. Canonical graphs checked and left unchanged: the IC scoreboard, `progress-timeline.json`, the leaderboard and `docs/bounds/FRONTIER.md` (no result here changes a frontier: every ratio to rho is above 1).